

Collaborative Scene Mood Authoring with Voice-driven Multimodal Feedback Design in Virtual Reality

Jinseok Hong*
KAIST UVR Lab
Daejeon, Republic of Korea
jindogliani@kaist.ac.kr

Minju Baeck*
KAIST UVR Lab
Daejeon, Republic of Korea
minjubaeck@kaist.ac.kr

Seokyoung Kim*
KAIST UVR Lab
Daejeon, Republic of Korea
seokyoung@kaist.ac.kr

Yoonseok Shin*
KAIST UVR Lab
Daejeon, Republic of Korea
sys7498@kaist.ac.kr

Woontack Woo†
KAIST KI-ITC ARRC/UVR Lab
Daejeon, Republic of Korea
wwoo@kaist.ac.kr



Figure 1: Our collaborative scene mood authoring system transforms the user’s voice command (input 1) into multimodal feedback (output) for the scene, leveraging LLMs along with scene metadata (input 2) and mood presets (input 3).

Abstract

We propose a collaborative scene mood authoring system in a virtual reality (VR) environment, where multiple users design and control the multimodal feedback (i.e., audiovisual and haptic) of a shared scene using voice commands. To support this, the user’s command is processed by large language models (LLMs) with the scene metadata and mood presets to identify intended target objects and mood, and the system modifies the objects to match the desired atmospheric expression. To cover a wide range of affective moods in scene mood authoring, we devised four distinct mood presets that represent different combinations of arousal and valence across the affective space. Each preset is expressed through a set of labeled visual, auditory, and haptic feedback elements, enabling coherent and multimodal expression of affective states within the scene.

*Co-First Authors: Authors contributed equally to this research.

†Corresponding Author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA XR '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2168-7/25/12

<https://doi.org/10.1145/3761667.3761960>

Keywords

Virtual Reality, Collaborative Authoring, Multimodal Feedback

ACM Reference Format:

Jinseok Hong, Minju Baeck, Seokyoung Kim, Yoonseok Shin, and Woontack Woo. 2025. Collaborative Scene Mood Authoring with Voice-driven Multimodal Feedback Design in Virtual Reality. In *SIGGRAPH Asia 2025 XR (SA XR '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3761667.3761960>

1 Introduction

In immersive virtual reality (VR) environments, mood plays a crucial role in shaping user experience, emotional engagement, and co-presence [Housand et al. 2024]. However, creating ambient moods remains a complex and unintuitive task, especially for non-experts. In-situ editing within VR adds further challenges, due to limited input affordances and increased cognitive load. Moreover, in multi-user collaboration, intuitive authoring becomes even more difficult due to increased coordination complexity. While recent advances have improved collaborative 3D scene editing, collaborative scene mood authoring is still constrained by a lack of intuitive controls. In parallel, with the advancement of large language models (LLMs), recent works in VR leverage natural language inputs (e.g., text and voice) as intuitive and explicit interfaces for authoring and design [Park et al. 2024]. Building upon these developments, we introduce an authoring system that enables natural voice-driven

control of scene mood through changes in multimodal feedback. By supporting real-time collaboration, our system allows users to collectively shape the mood of a shared scene through synchronized multimodal feedback—much like having a conversation.

2 Collaborative Scene Authoring

To facilitate interaction events and virtual objects synchronization among users, we built a **collaborative authoring environment** via Photon coupled with Unity's Netcode framework. Interaction events (*i.e.*, editing objects via voice commands) are propagated in real-time through the network. Interaction events executed in the authoring environment are streamed to all connected clients, which receive and apply these events in the order they are issued, ensuring that updates are sequentially overwritten and reflected in the shared scene. To support this, each object contains synchronization state metadata. By doing so, the system ensures consistent updates and natural interactions in collaborative authoring.

In our system, users modify the multimodal feedback of the shared scene through **voice commands**. To support this interaction, we leverage the Meta XR Voice SDK to convert voice commands into a textual format via speech-to-text (STT) processing. The converted text is then transmitted to LLMs (GPT-4o via OpenAI API in the demo) along with prompts via Unity WebRequest. The LLMs identify which objects within the scene the user's input refers to and determine which of the four mood presets defined in our multimodal feedback design most semantically matches the mood expressed by the user. To this end, the prompt incorporates metadata of the scene and the mood presets. Based on the LLM output, the system updates the target object by selecting the relevant audiovisual and haptic elements labeled under the matched preset, thereby modifying its expression within the scene. In this demo, the system runs on Meta Quest 3, utilizing its built-in microphone to capture voice commands and delivering audiovisual feedback directly through the headset's integrated display and speakers.

3 Multimodal Feedback Design

We devised four distinct **mood presets** (*Euphoria*, *Spicy*, *Blue*, and *Chill*) by adapting Russell's circumplex model of affect [Russell 1980] to cover a wide range of affective moods in our scene mood authoring system. We adapted representative affective states more towards the moods of spaces and objects. Each preset corresponds to different combinations of arousal and valence levels across the four quadrants of the affective space. **Euphoria** (Q1: high arousal, high valence) radiates a vibrant emotional state filled with optimistic, uplifting, and playful energy. **Spicy** (Q2: high arousal, low valence) channels a raw and provocative atmosphere driven by tension or emotional volatility, evoking dramatic catharsis and rebellious energy. **Blue** (Q3: low arousal, low valence) reflects a deeply resonant and contemplative mood, characterized by emotional stillness and quiet introspection. **Chill** (Q4: low Arousal, high valence) evokes a peaceful and calm state, marked by serenity and balance.

Once the mood presets and target objects are determined, the system retrieves and applies the corresponding predefined **affective multimodal feedback** elements (visual color themes, ambient sound, and haptic feedback) from the database. The system supports global mood changes of multimodal feedback elements, as well as object-level changes, allowing users to modify the feedback

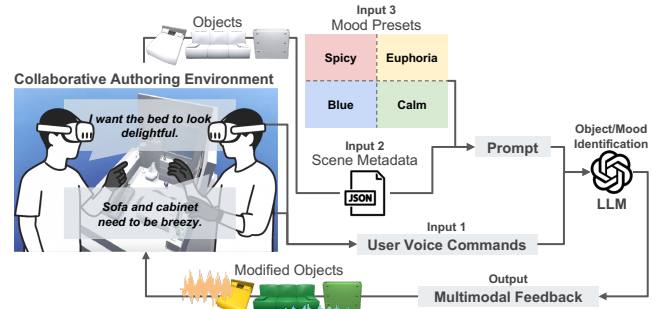


Figure 2: Scene mood authoring system diagram

of individual objects. **Visual**. Color themes are used to represent ambient moods based on affective dimensions. We referenced prior work on affective color mapping in visualization [Bartram et al. 2017]. **Haptic**. Based on the research of [Baeck et al. 2025], we used the bHaptics TactGlove DK2, adjusting vibration intensity to amplify the arousal dimension along with the overall audiovisual mood, thereby shaping the integrated multimodal scene expression. **Audio**. Ambient sound is used to support the overall mood of the scene. To ensure that the ambient sound follows affective states of the mood presets, we utilized the 4Q Audio Emotion Dataset [Panda et al. 2018], which follows Russell's circumplex model of affect.

4 Conclusion and Future Work

We introduced a collaborative scene mood authoring system using voice commands to design the multimodal feedback of scenes. For future demonstrations, we plan to enrich the set of feedback elements to support a more expressive and diverse multimodal feedback design by incorporating generative AI models. For future avenues, integrating 3D scene synthesis techniques such as lighting, layout design, and object placement into our system may enable comprehensive and extended scene mood authoring.

Acknowledgments

This work was supported by the IITP (Institute of Information & Communications Technology Planning & Evaluation)-ITRC (Information Technology Research Center) grant (IITP-2025-RS-2024-00436398) and the metaverse support program to nurture the best talents (IITP-2025-RS-2022-00156435) funded by the Korea government (Ministry of Science and ICT).

References

- Minju Baeck, Yoonseok Shin, Dooyoung Kim, Hyunjin Lee, Sang Ho Yoon, and Woon-tack Woo. 2025. Visuo-Tactile Feedback with Hand Outline Styles for Modulating Affective Roughness Perception. *arXiv preprint arXiv:2508.13504* (2025).
- Lyn Bartram, Abhisekh Patra, and Maureen Stone. 2017. Affective color in visualization. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 1364–1374.
- Jessica Housand, Allen Cornelius, and Karen E Shackleford. 2024. Greater usage and positive mood change for users of a dynamic VR app before and after the COVID-19 pandemic onset. *Frontiers in Psychology* 15 (2024), 1278207.
- Renato Panda, Ricardo Malheiro, and Rui Pedro Paiva. 2018. Novel audio features for music emotion recognition. *IEEE Transactions on Affective Computing* 11, 4 (2018), 614–626.
- Hyerim Park, Jinseok Hong, Heejeong Ko, and Woontack Woo. 2024. Revolutionizing Metaverse Content Creation: GenAI's Role and Future Directions. *IEEE Consumer Electronics Magazine* (2024), 1–7. doi:10.1109/MCE.2024.3501512
- James A. Russell. 1980. The circumplex model of affect. *Journal of Personality and Social Psychology* 39, 6 (1980), 1161–1178.